DNA-Based Disease Prediction

Abstract

Recent advances in genomics have crossed the limits of biotechnology. In today's era, many solutions to deadly diseases are derived from genetic advancements. As we know, DNA stores entire data — such as ancestry, traits, immunity, and proteins — in an encoded form. This project focuses on developing a computational model that analyzes genomic variations, such as Single Nucleotide Polymorphisms (SNPs), to predict disease scores.

Several ML algorithms — Linear Regression and Random Forest — were applied to train the model to generate the risk label of the respective disease. The model uses Genome-Wide Association Studies (GWAS) datasets and Variant Call Files (VCFs) of patients to identify risk alleles associated with diseases.

As input, we accept patient SNP IDs with genotype data and chromosome numbers; considering this, we predict the Polygenic Risk Score of a disease. The system integrates bioinformatics and machine learning techniques to provide predictive insights that may assist in early diagnosis and precision medicine.

The system accurately predict Polygenic Risk (PRS) for disease with the accuracy of 85% across datasets . These statistics describe its reliability in the practical world.

1. Introduction

Every human has a unique DNA sequence that contains genetic information determining various biological traits, including disease susceptibility. However, a gene may be common among different people but can produce a faulty protein, causing an individual to suffer from a disease. There might also be a gene located in the wrong position or carrying a different allele, which could lead to many diseases. In the last decade, the availability of genome sequencing data and the rise of genetic engineering have made it possible to analyze genetic markers and correlate them with diseases.

However, most of the available data is fragmented and too large for researchers and doctors to analyze efficiently. Moreover, for a genetic disease, multiple genes can be responsible, so analyzing a single gene is not sufficient. We need to analyze multiple factors, which is not feasible for an individual. Therefore, there is a need for a system that can analyze a patient's VCF file and extract all potential risks related to genetic disorders.

This System may also help identify new SNP–disease associations or validate existing ones across different populations.

2. Motivation

Genetic sequences are extremely vast , containing millions or even billions of base pairs . Most of these patterns may not be directly relevant, making manual analysis impractical.

Why does this project matter ?

1. **Human analysis or traditional computing methods are insufficient**, as the volume and complexity of genomic data can easily lead to misinterpretation, errors, or incomplete insights.
2. **There is a need for an intelligent system that can efficiently analyze genomic data**, identify meaningful patterns, and provide probabilistic predictions about potential genetic risks—even those that may affect future generations.
3. **This project helps promote awareness about an individual's genetic makeup**, enabling better understanding of inherited traits, health risks, and personalized medical insights.

3. Objectives

- To analyze genetic variations (SNPs) from DNA data.
- To integrate GWAS dataset with VCF files for gene disease mapping.
- Develop an efficient system for predicting the Polygenic Risk Scores (PRS) of the Diseases.
- To provide a user-friendly interface (UI) that enables researchers, clinicians, and patients to interact effectively with the prediction model.
- Traits Analyzer of any Human and Animal **. [Future]**

4. Literature Survey

A considerable amount of Studies has been conducted to understand how genomic variations influence human health and disease and explored the association or co-relation between genomics and Disease and Several Researches are exploring about DNA modification , Disease Idendification Genes :

- **NHBI (National Human Genome Research Institute)**
  NHBI programs provide vast database of clinically relevant genetic variants. These Resource help in developing computational models.

- **Genome-Wide Association Studies (GWAS)**
  GWAS are statistical studies that provides information about SNPIDs with beta score and associated Disease.

- **1000 Genome Project**
  This Project produced over 80 millions genetic variants , including SNPs , reference panels for genotype imputation and population frequency data.

- **IEEE Xplore: DNA-Based Disease Prediction Using Pathway Analysis**
  This Research paper provides methods (ML models or DL models) like SVM , Random Forest and Neural Network to identify Disease.

5. Dataset

Model Trained on following Datasets :

A. **GWAS catalog**
   The Dataset includes SNPIDs along with their respective effct sizes , p – values and associated Disease.
B. **Patient Data**
   The Dataset includes patient's SNPIDs and their Genotype with PRS

Attribute Extracted : rsId , chromosome (CHR) , position (POS) , reference allele (REF) , alternate allele (ALT) , genotype and associated disease trait.

Data Preprocessing :

The following steps were used for data cleaning and transformation :

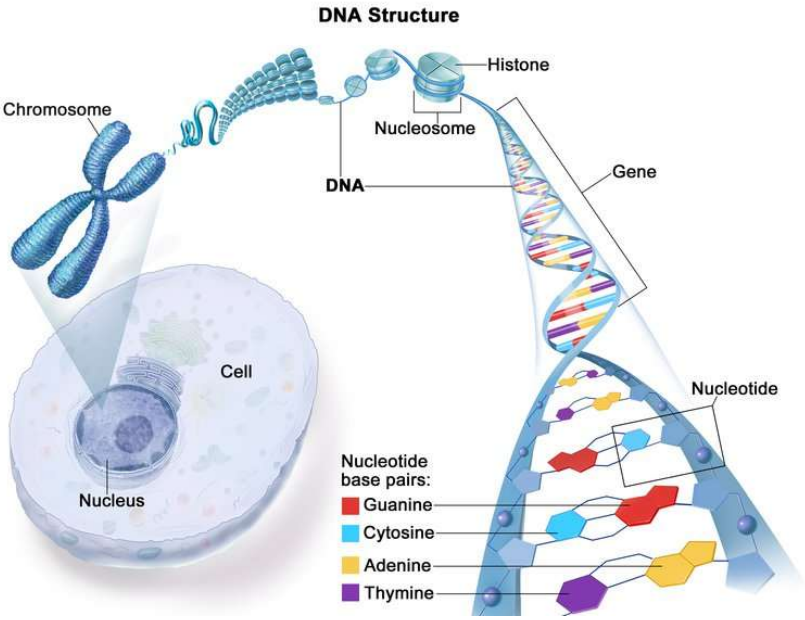
1. Remove Redundant columns
2. Missing Value Handling
3. SNP to Disease Mapping
4. Encoding genotypes into numerical format (0 , 1 , 2) for ML model.

6. Solution Framework

6.1 DNA Structure

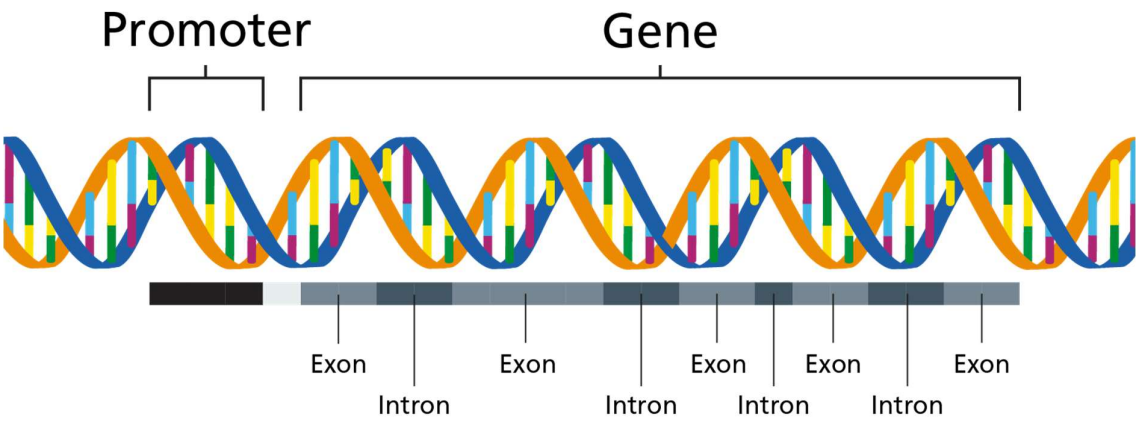
DNA (Deoxyribonucleic Acid) is the hereditary material found in cells . It made of nucleotides : Adenine (A) , Thymine (T) , Cytosine (C) and Guanine (G) . These Nucleotides are arranged in a double – helix structure.

DNA is organized into chromosomes and each chromosome contains thousands of genes.



6.2 How a Gene Affects the Body

AS we know that gene is a segment of DNA that contains instructions to make proteins which control all biological processes such as metabolism , immunity , growth etc. But all genes are beneficial , that's not true because some genes produce such proteins that harm out immune , or if a gene is muted , the protein may not function well.



6.3 Useful Data Extracted from DNA

Patients.csv : **(id , rsid , chromosome , pos , genotype , PRS)**

GWAS.csv : (DISEASE,SNP,RISK_ALLELE,BETA,CHR)

Here we observe 3 attributes about data that can be a deciding factor of disease prediction :

A. **Effect Size (OR / BETA) :**

This measures how strongly a genetic variant (SNP) is associated with a trait or disease. There are 2 common measures –

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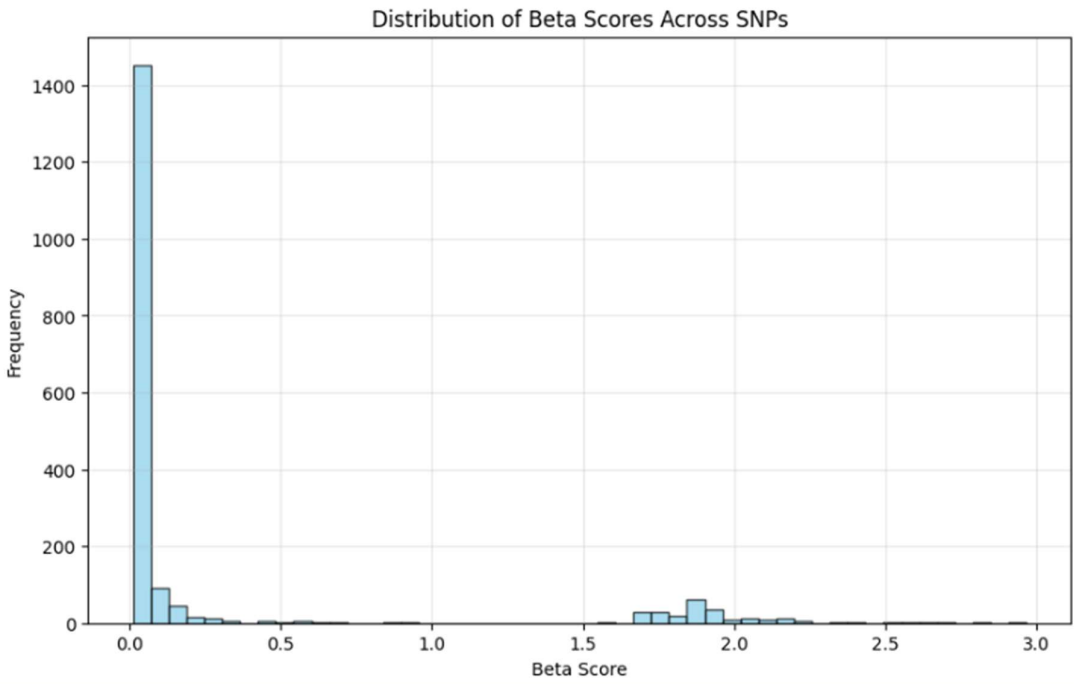
i. Odds Raito : This measures how much the SNP increases or decreases Probability of risk . This used for Binary Classification.

OR = 1 => SNP has no effect
OR > 1 => SNP increases disease risk
OR < 1 => SNP decreases disease risk

ii. Beta (β) : This measure used for Continuous and quantitative outcomes such as physical traits or blood pressure etc. This represent how much a SNP change the trait.

β is positive => SNP increases trait value
β is Negative => SNP decreases trait value

Beta Value Distribution



6.4 Prediction Formulae

6.4.1 PRS (Polygenic Risk Score) Calculation

PRS quantifies the overall genetic risk of a disease by combining the effect of multiple SNPs. Each SNP holds two values genotype and risk allele that combinaly with beta score contributes to calculate PRS value.

PRS for one SNP = (Number of risk alleles in genotype) × Beta

6.4.2 Random Forest Classifier

Random Forest is ensemble of decision Trees. Each tree predicts the outcome and majority of one class is predicted as final outcome.

Each Tree T_j predicts outcome y_j then we compute the final outcome by applying mode operation on all outputs.

$$\boldsymbol{Y_{RF} = mode(y_1 , y_2 , y_3 , \dots \dots \dots \ , y_n)}$$

6.4.3 Gradient Boosting Classifier

Gradient Boosting Classifier is also a ensemble method of decision tree where each tree is joined sequentially , and each tree tries to correct the errors of previous tree . It minimizes a loss function .

6.4.4 Support Vector Machine (SVM)

SVM generates a hyperplane that efficiently separates the classes by maximizing the margin between support vectors. Our model uses three output labels (Low , Medium and High) so we have to use support vector kernel that efficiently claasied each class in a 3d plane .
This method is effective in high – dimensional feature space.

6.4.5 Logistic Regression

Logistic Regression models the probability of a categorical outcome with the help of sigmoid function.

$$\boldsymbol{Risk\ Score = B_0 + B_1 x_1 + B_2 x_2 + \ \dots\dots + b_n x_n}$$

X_i : Genotype value (0 / 1 / 2)
B_i : weight / effect size

B_0 : bias / intercept

Sigmoid Function :

$$P(y) = \frac{1}{1 + e\text{^}(-(Risk\ Score))}$$

7. Methodology

To Train the model with the GWAS and Patients dataset , we follows several Process through which our model train and predict results.

7.1 Data Cleaning & Preprocessing

- First we remove rows that does not contain SNPIDs
- Encoding Genotype (with 0 / 1 / 2)
- Genotype Cleaning : Filled missing genotype values with 'NN'.
- Mapping of SNPs to Disease
- Feature are scaled using standard Scalar

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7.3 Feature Selection

After Preprocessing the data , we have to select the useful featres By which we can train a model and which captures both biological significance and genetic variation.

Features are :

- rsId : SNP Identifier
- chromosome : chromosome Number
- pos : SNP position on chromosome
- genotype : numeric genotype
- BETA : effect size of SNP

7.4 Model Training

Model trained on four machine learning models tp classify PRS risk groups :

1. Random Forest
2. Gradient Boosting
3. Support Vector Machine
4. Logistic Regression

We split the data into three sections Training (60%) , Testing (20%) and Validation (20%) , validation data used to overcome overfitting the data.

7.5 Model Evaluation

To evaluate the model we used confusion metrics and it's parameters – Acuuracy , Precision , Recall , F1 – Score , and RMSE (Root mean Squared Error)

8. Results and Discussion

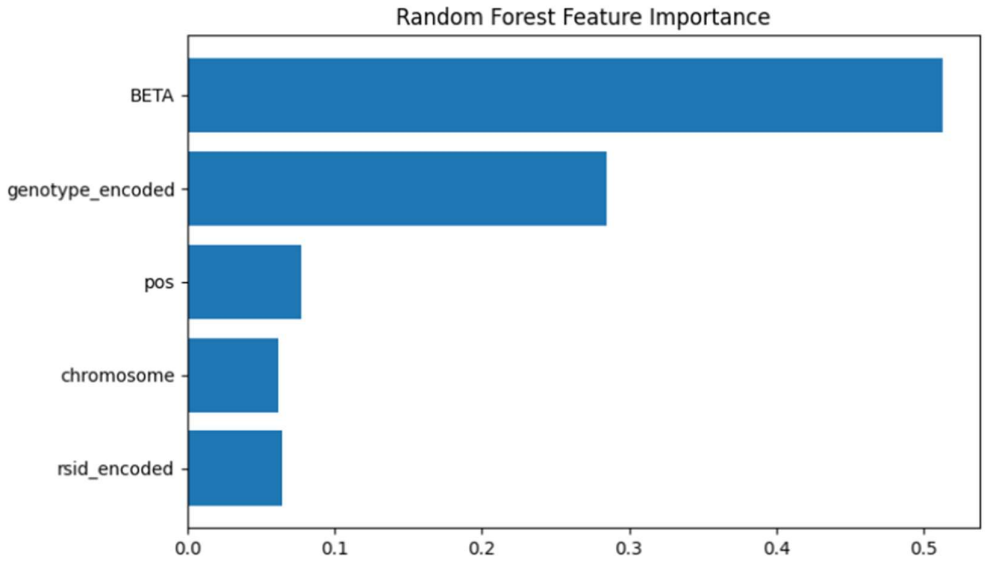
- Gradient Boosting and Random Forest achieved the maximum accuracy respectively are : 93% and 92% .
- The Syatem Successfully identified the Risk Score (Low / Medium / High) with max beta score and associate Disease prediction.

Validation Results:

	Model	Accuracy	Precision	Recall	F1-Score	RMSE
0	Random Forest	0.9225	0.920654	0.9225	0.921114	0.316228
1	Gradient Boosting	0.9300	0.931788	0.9300	0.930757	0.304138
2	SVM	0.8175	0.746368	0.8175	0.778346	0.427200
3	Logistic Regression	0.7900	0.725228	0.7900	0.755753	0.458258

Test Results:

	Model	Accuracy	Precision	Recall	F1-Score	RMSE
0	Random Forest	0.9450	0.945110	0.9450	0.945019	0.234521
1	Gradient Boosting	0.9500	0.950690	0.9500	0.950202	0.239792
2	SVM	0.8075	0.729482	0.8075	0.764324	0.438748
3	Logistic Regression	0.7700	0.795912	0.7700	0.735405	0.479583



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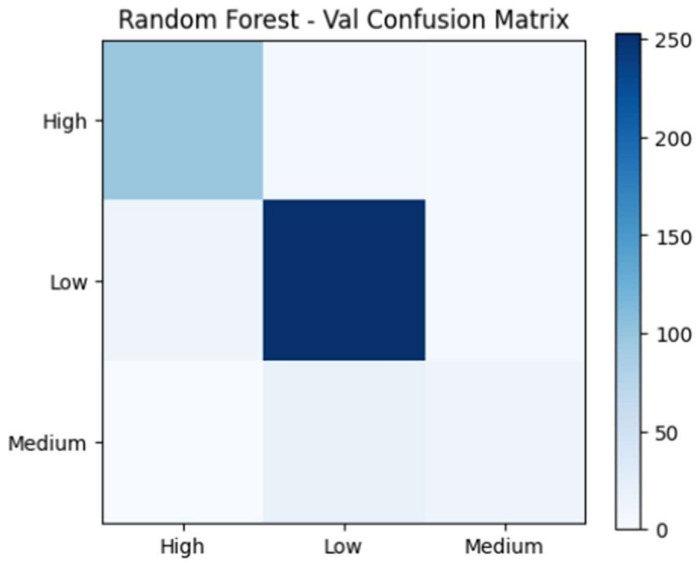

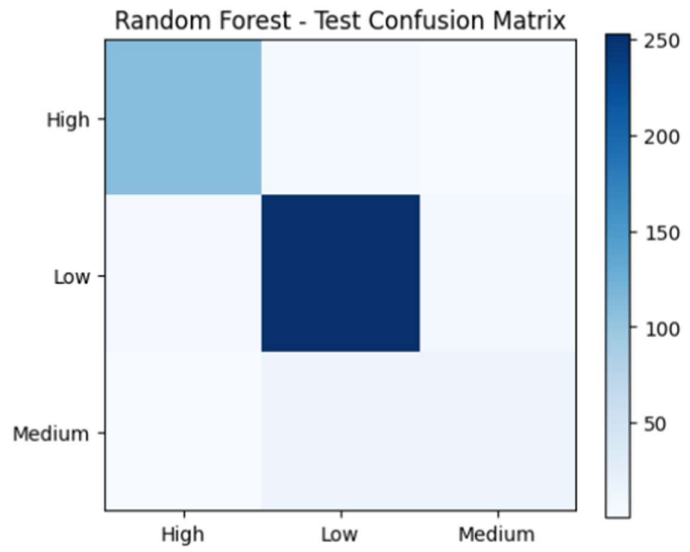

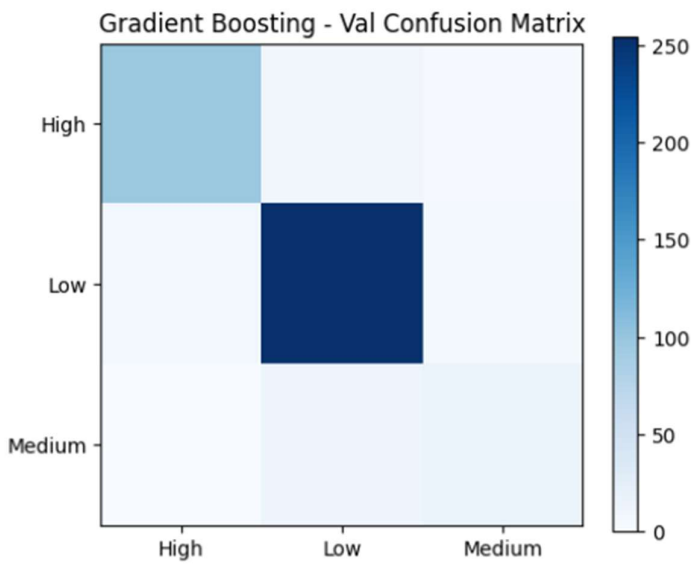

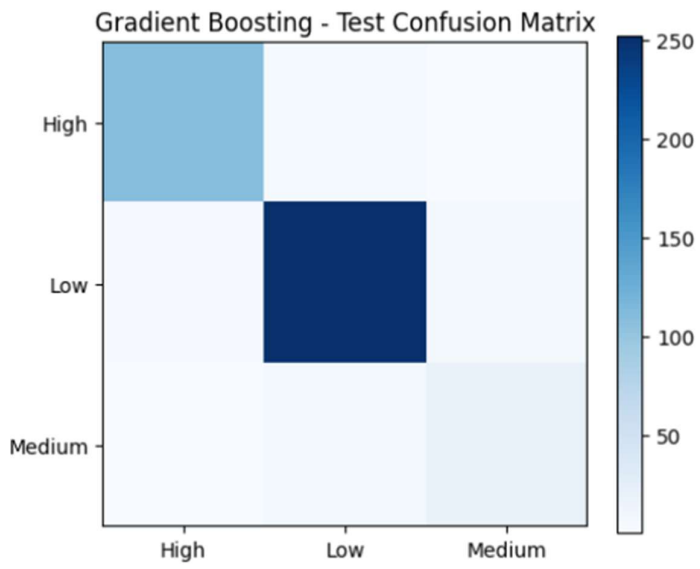

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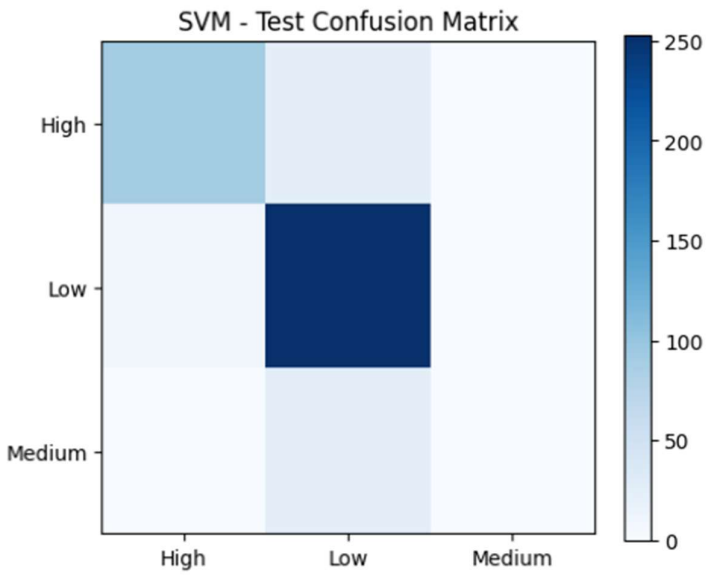
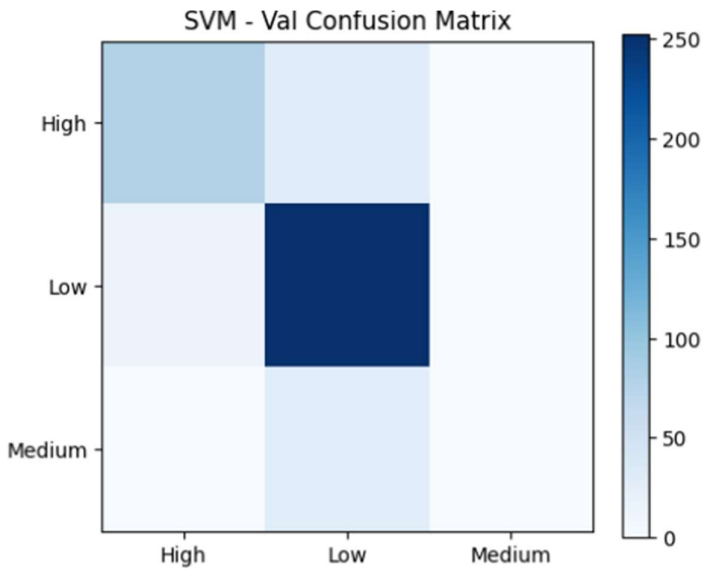
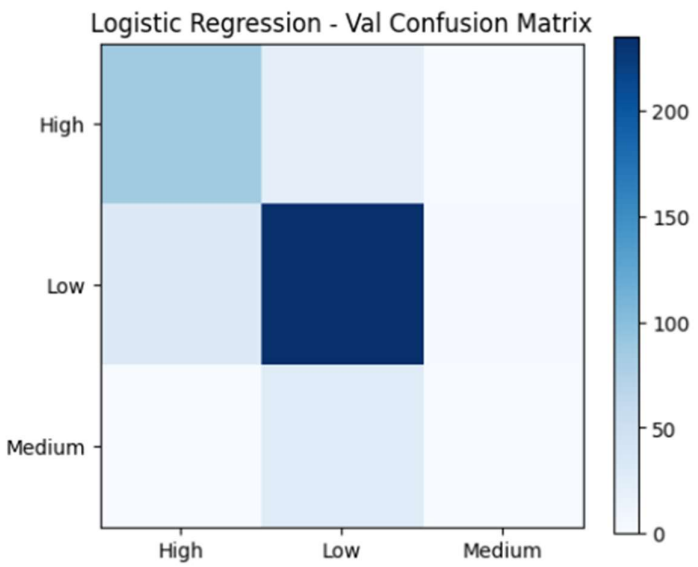
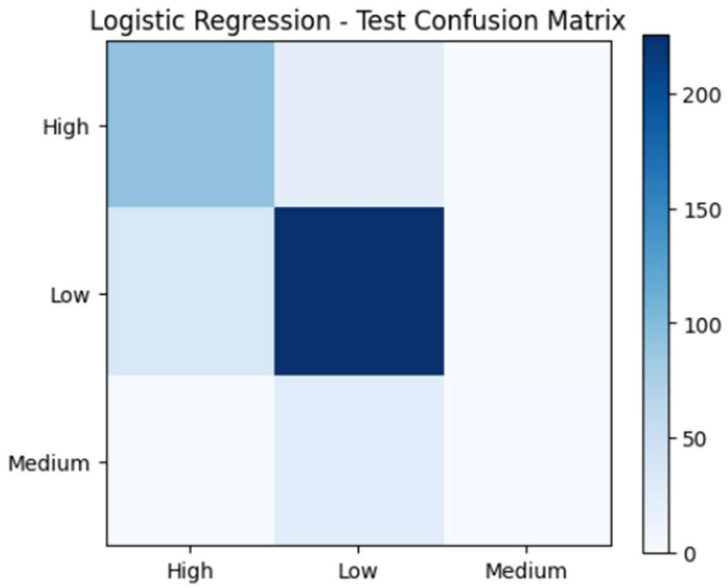
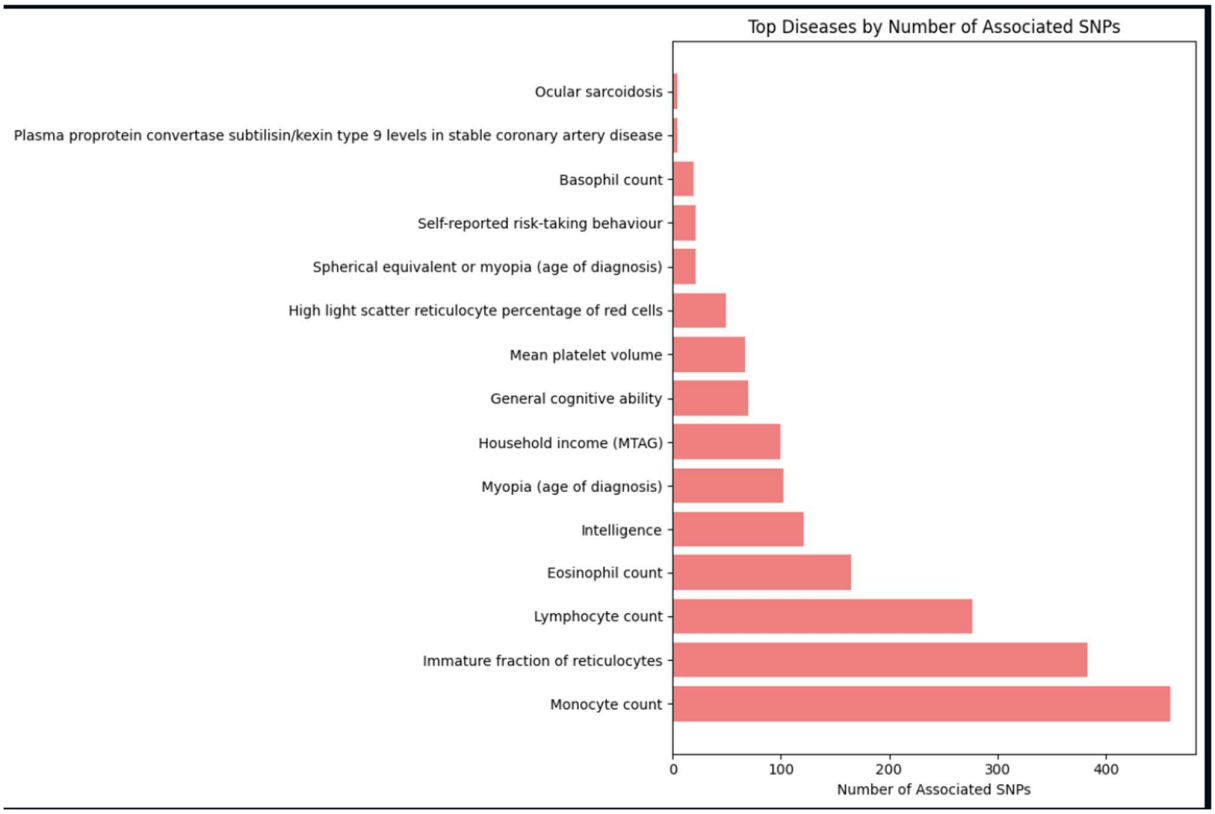








Several Diseases Used in DB with SNP



9. Conclusion

This project demonstrates a data driven approach to predict disease risks using DNA information. By analyzing SNP and genotype variations and their associated disease , the system provides a foundation for genetic diagnostics and personalized medicine.

Future work may include creating a UI framework for end user , expanding dataset , applying Deep Learning Techniques to identify hidden along with build a new possibility and integrating clinical data.

10. References

1. GWAS Catalog – https://www.ebi.ac.uk/gwas/
2. 1000 Genomes Project – https://www.internationalgenome.org/
3. IEEE Xplore - https://ieeexplore.ieee.org/document/7976867
4. Scikit-learn Documentation – https://scikit-learn.org/